GATeR: Graph-Aware Test Repair

Project Team

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Chapter 1

Introduction

This chapter sets the stage by providing an overview of the study's focus and objectives. It outlines the problem domain, research problem statement, and software requirements specification to give readers a comprehensive understanding of the study's scope and purpose.

1.1 Problem Domain

Automated test case repair is a critical challenge in software maintenance and quality assurance. When software systems evolve, test cases frequently break due to changes in the System Under Test (SUT), leading to significant maintenance costs and development delays. Research indicates that broken test cases account for 14% to 22% of test failures in open-source projects, disrupting continuous integration pipelines and wasting valuable developer time.

Current automated test repair approaches suffer from "contextual blindness"—they focus solely on immediate code changes while missing the rich contextual information available in the repository ecosystem. Existing solutions like TARGET [11] achieve 66.1% exact match accuracy but lack holistic understanding of project-specific patterns, documentation, historical changes, and code relationships. This limitation prevents generation of truly idiomatic and maintainable test repairs that align with a project's unique conventions.

The field encompasses several key challenges: (1) resource-intensive graph construction taking 45-60 minutes for large codebases, (2) limited context windows in Large Language Models (LLMs) causing token overflow, (3) high memory consumption of 4-6GB for traditional approaches, and (4) lack of repository-level understanding including documentation, issues, pull requests, and code dependencies.

1.2 Research Problem Statement

How can we develop an enterprise-ready, automated test repair system that generates idiomatic and maintainable test repairs by efficiently and intelligently leveraging comprehensive, repository-level contextual knowledge, while overcoming limitations of contextual blindness, resource intensity, and LLM context constraints?

Current test repair tools exhibit the following critical limitations:

- **Incomplete Contextual Understanding:** Existing tools miss project-specific patterns, utility functions, coding conventions, documentation (READMEs), and historical context from GitHub issues and pull requests, preventing generation of idiomatic repairs.
- **Resource Intensiveness:** Building comprehensive knowledge graphs for large codebases (500,000 lines of code) using traditional methods takes 45-60 minutes and consumes 4-6GB memory, making frequent local use impractical.
- **LLM Context Window Limitations:** Limited context windows (4K-32K tokens) in LLMs cause token overflow, reduced efficiency, and decreased accuracy when processing vast repository information.
- **Lack of Interpretable Reasoning:** Many LLM-based approaches lack transparent reasoning chains for repair decisions, limiting trustworthiness and practical adoption.

1.3 Software Requirements Specification

1.3.1 Functional Requirements

FR1: GitHub Repository Access

- Support GitHub repository access via API
- Fetch repository metadata including issues, pull requests, and commit history
- Implement authentication and rate limiting for API access
- Enable local caching for offline analysis after initial retrieval

FR2: Intelligent Knowledge Graph Construction

- Parse code using Tree-sitter for efficient incremental processing
- Build comprehensive knowledge graphs linking code, documentation, issues, and pull requests
- Support codebase entities (files, classes, functions) and repository artifacts (issues, PRs)
- Enable rapid setup for large codebases

FR3: Advanced Context Retrieval

- Implement multi-hop graph traversal for relationship discovery
- Perform semantic similarity search using vector embeddings
- Calculate relevance scores using KGCompass methodology
- Combine graph-based and semantic search results

FR4: Test Repair Generation

- Generate idiomatic test repairs using LLM with augmented context
- Support multiple programming languages (Java, Python initially)
- Validate generated repairs for syntax correctness
- Aim to surpass existing automated test repair tools in accuracy and reliability

FR5: Performance Optimization

- Implement multi-level caching (memory, disk, compressed)
- Use incremental parsing to minimize redundant work
- Maintain efficient memory usage for practical deployment
- Enable rapid repair generation through intelligent caching

1.3.2 Non-Functional Requirements

NFR1: Privacy and Security

- All processing performed locally on developer's machine
- No code transmitted to external servers
- Support offline operation without internet connectivity

NFR2: Scalability

- Handle repositories from 10K to 1M+ lines of code
- Sub-second incremental updates for code changes
- Efficient batch processing for multiple broken tests

NFR3: Usability

- Command-line interface for developer workflow integration
- VS Code extension for IDE integration (optional)
- Web service API for team collaboration (optional)
- No complex setup or external dependencies required

NFR4: Maintainability

- Modular architecture with clear separation of concerns
- Comprehensive logging for debugging
- Extensible design for additional languages and features

NFR5: Performance

- Rapid initial setup for large codebases
- Efficient memory usage for local deployment
- Fast context retrieval through optimized indexing
- Quick repair generation with intelligent caching
- High cache efficiency after initialization

Chapter 2

Literature Review

This chapter critically examines existing literature relevant to the research topic. It identifies key studies, theories, and findings, providing a foundation for the proposed research and highlighting its relationship to prior work.

2.1 Related Research

This section examines existing approaches in automated test repair and related technologies, analyzing their strengths, weaknesses, and relationship to GATeR.

2.1.1 TARGET (Test Repair Generator)

2.1.1.1 Summary of the research item

TARGET [11], developed by Yaraghi et al. (2024), represents a significant advancement in automated test repair using pre-trained Code Language Models (CLMs). The approach treats test repair as a language translation task, fine-tuning models like CodeT5+ [9] (770M parameters) on a comprehensive benchmark called TARBENCH containing 45,373 broken test repairs across 59 projects.

TARGET achieves 66.1% exact match accuracy and 80% plausible repair accuracy through a two-step process: (1) repair context identification and prioritization using static code analysis and call graphs, and (2) CLM fine-tuning with multiple input-output formats. The system identifies method-level and class-level code changes (hunks) in the SUT, prioritizes them using call graph depth, context type, and TF-IDF similarity, then formats this information for CLM consumption. TARGET supports both line-level and word-level hunk representations, exploring four different input-output formats to optimize repair generation.

2.1.1.2 Critical analysis of the research item

Strengths:

- Comprehensive benchmark (TARBENCH) with rigorous validation through three-phase test execution
- Language-agnostic design applicable beyond Java
- Superior performance compared to baselines (37.4 percentage points improvement over non-context approaches)
- Fully reproducible with open-source code and datasets
- Addresses multiple repair categories (ARG, INV, ORC) without restriction to specific types

Weaknesses:

- Limited to single-hunk test repairs, excluding multi-hunk scenarios
- Relies solely on SUT code changes, missing broader repository context (documentation, issues, unchanged code)
- Call graph generation through static analysis can be imprecise
- Context selection limited to 512 tokens due to computational constraints
- No integration of repository artifacts like GitHub issues or pull requests
- Performance degrades with repair complexity (20% EM for complex multi-category repairs)

2.1.1.3 Relationship to the proposed research work

TARGET establishes the foundation for LLM-based test repair and provides the benchmark for comparison. GATeR addresses TARGET's primary limitation: contextual blindness. While TARGET focuses on immediate code changes within token limits, GATeR incorporates comprehensive repository-level context through knowledge graphs, semantic search, and multi-hop reasoning. GATeR introduces improved context-aware capabilities through KGCompass relevance scoring, LanceDB vector search, and GraphRAG context management, with the aim of surpassing existing tools in accuracy and contextual understanding.

2.1.2 CEPROT (Test Update Generation)

2.1.2.1 Summary of the research item

CEPROT [6] (Hu et al., 2023) addresses automatic detection and updating of obsolete tests by fine-tuning the CodeT5 language model [9]. The approach defines obsolete tests as those modified within a commit where the corresponding production method also changes, matching tests to production methods via file paths and method names.

CEPROT's dataset contains 5,196 instances with 520 evaluation instances. The system generates full test code as output and represents SUT changes by including both versions of the full production method along with edit sequences. CEPROT achieves 12.3% exact match accuracy and 63.1% CodeBLEU score, focusing exclusively on co-evolving production-test pairs within single commits.

2.1.2.2 Critical analysis of the research item

Strengths:

- Addresses both detection and repair of obsolete tests
- Uses edit sequence encoding for representing changes
- Attempts to capture production-test co-evolution

Weaknesses:

- Very limited dataset diversity (4,676 training instances)
- Narrow definition of test repairs (only co-evolving production-test pairs)
- Doesn't verify repair success through test execution
- Includes trivial repairs (17% of instances) inflating results
- Random train-test split doesn't prevent temporal data leakage
- Restrictive 150-token limit for input and output
- Focuses only on production method changes, missing broader context
- Contains data quality issues (11% mismatched commits, 5% identical source/target, 7% mismatched method names)
- Poor performance (12.3% EM) compared to state-of-the-art

2.1.2.3 Relationship to the proposed research work

CEPROT demonstrates the limitations of narrow context approaches in test repair. Its poor performance (12.3% EM vs. TARGET's 66.1%) highlights the critical importance of comprehensive context. GATeR addresses CEPROT's fundamental weaknesses by: (1) using rigorous test execution validation, (2) incorporating repository-wide context beyond production methods, (3) eliminating trivial repairs from evaluation, (4) supporting larger context windows (512 tokens), (5) training on substantially larger datasets, and (6) implementing proper temporal train-test splitting.

2.1.3 KGCompass (Knowledge Graph for Program Repair)

2.1.3.1 Summary of the research item

KGCompass [10], proposed by Yang et al. (2025), introduces a repository-level software repair framework using knowledge graphs to connect repository artifacts (issues, pull requests) with codebase entities (files, classes, functions). The system constructs a repository-aware knowledge graph and uses multi-hop traversals to narrow search spaces to the most relevant functions.

KGCompass's key innovation is its relevance scoring formula that combines structural proximity (path length), semantic similarity (embedding cosine similarity), and textual similarity (Levenshtein distance) with configurable weights. The research demonstrates that 69.7% of successfully localized bugs required multi-hop traversals, proving relationships cannot be captured by direct textual matching alone. KGCompass achieves 45.67% repair performance and 51.33% function-level localization accuracy, representing state-of-the-art in repository-level program repair.

2.1.3.2 Critical analysis of the research item

Strengths:

- Pioneering use of knowledge graphs for repository-level understanding
- Empirically validated multi-hop reasoning importance (69.7% of fixes)
- Balanced relevance scoring combining multiple similarity measures
- Demonstrates value of connecting artifacts with code entities
- Provides explanations for generated patches through graph traversal
- Addresses the "vast search space" problem in program repair

Weaknesses:

- Focused on program repair (buggy code) rather than test repair
- Limited scalability information for very large repositories
- Graph construction time not thoroughly optimized
- Dependency on graph database infrastructure
- Relevance formula parameters may require tuning per project

2.1.3.3 Relationship to the proposed research work

KGCompass provides the foundational methodology for GATeR's knowledge graph construction and relevance scoring. GATeR directly adapts KGCompass's: (1) graph schema for linking artifacts and code entities, (2) relevance scoring formula for prioritizing context, (3) multi-hop traversal strategy for discovering relationships, and (4) emphasis on structural proximity over pure textual matching. However, GATeR extends KGCompass by: (1) applying the methodology to test repair instead of program repair, (2) optimizing for local embedded databases (Kùzu) rather than infrastructure-heavy solutions, (3) integrating with vector search (LanceDB) for enhanced semantic understanding, and (4) combining with GraphRAG for intelligent context augmentation and LLM integration.

2.1.4 DSrepair (Knowledge-Enhanced Data Science Program Repair)

2.1.4.1 Summary of the research item

DSrepair [8] (Ouyang et al., 2025) introduces knowledge-enhanced program repair specifically for data science code, leveraging knowledge graph-based Retrieval-Augmented Generation (RAG) [7] and bug information enrichment. The system builds DS-KG, a knowledge graph for data science libraries, and uses AST-node level bug information to localize errors at fine-grained granularity.

DSrepair's approach combines: (1) domain-specific knowledge graphs capturing API usage patterns and library relationships, (2) AST-level error localization for precise bug identification, (3) RAG pipeline to augment LLM context with relevant library documentation and usage examples, and (4) fine-grained bug information enrichment. The system demonstrates significant improvements in repairing data science code, particularly for library API usage errors.

2.1.4.2 Critical analysis of the research item

Strengths:

- Domain-specific knowledge graphs tailored for data science libraries
- Fine-grained AST-level error localization
- Effective RAG integration for library documentation retrieval
- Addresses specific challenges in data science code repair
- Demonstrates value of knowledge enrichment for LLM-based repair

Weaknesses:

- Limited to data science domain and specific libraries
- Scalability to general-purpose programming unclear
- Knowledge graph construction requires domain expertise
- Evaluation limited to specific library error types
- Unclear performance on non-library-related errors

2.1.4.3 Relationship to the proposed research work

DSrepair validates the power of combining knowledge graphs with RAG [7] for code repair tasks. GATeR adopts DSrepair's core insight—that augmenting LLMs with structured knowledge dramatically improves repair quality—but applies it to test repair with broader scope. GATeR generalizes DSrepair's approach by: (1) building repository-wide knowledge graphs instead of library-specific ones, (2) incorporating multiple knowledge sources (code, documentation, issues, PRs) beyond API documentation, (3) supporting general test repair rather than domain-specific bug fixes, and (4) implementing more sophisticated context selection through KGCompass scoring.

2.1.5 Tree-sitter (Incremental Parsing)

2.1.5.1 Summary of the research item

Tree-sitter [3] is an incremental parsing system designed for code editors, supporting multiple programming languages with error recovery and minimal reparsing. Unlike traditional parsers that reprocess entire files on each change, Tree-sitter maintains parse tree state and only reparses modified sections, achieving dramatic performance improvements.

Tree-sitter's architecture includes: (1) incremental parsing that reuses unchanged parse tree nodes, (2) error recovery allowing partial parsing despite syntax errors, (3) language-agnostic design supporting multiple languages through grammar definitions, (4) query system for extracting specific patterns from ASTs, and (5) binding support for multiple programming languages. The system is production-proven in editors like Atom, Neovim, and Emacs, demonstrating real-world reliability and performance.

2.1.5.2 Critical analysis of the research item

Strengths:

- Proven 85% performance improvement over traditional parsers
- Robust error recovery unlike traditional parsers that fail on syntax errors
- Language-agnostic design enabling multi-language support
- Production-proven in major code editors
- Memory-efficient through parse tree reuse
- Well-maintained with active community support

Weaknesses:

- Requires grammar definitions for each language
- Limited built-in semantic analysis
- Some advanced language features challenging to represent
- Documentation could be more comprehensive for advanced use cases

2.1.5.3 Relationship to the proposed research work

Tree-sitter serves as GATeR's parsing engine, providing significant performance improvements critical for practical deployment. GATeR leverages Tree-sitter's: (1) incremental parsing for efficient code updates during development, (2) error recovery for handling broken test code gracefully, (3) multi-language support for extensibility beyond Java, (4) query system for extracting code entities (classes, methods, tests), and (5) memory efficiency for maintaining reasonable resource usage.

2.1.6 GraphRAG (Graph-Based Retrieval-Augmented Generation)

2.1.6.1 Summary of the research item

GraphRAG [5], proposed by Han et al. (2025), represents a significant evolution of traditional Retrieval-Augmented Generation (RAG) by incorporating graph-structured data to enhance contextual understanding and reasoning in large language models. While conventional RAG methods rely on retrieving semantically relevant textual information, GraphRAG leverages graph-based retrieval mechanisms to capture complex relational and structural information unavailable in linear text.

The holistic GraphRAG framework comprises five core components: (1) Query Processor that translates user queries into graph-compatible formats through entity recognition and relation extraction, (2) Retriever that identifies relevant graph content using heuristic, learning-based, or hybrid neural-symbolic methods, (3) Organizer that refines retrieved graph data through pruning, reranking, and augmentation, (4) Generator that integrates refined graph knowledge into LLM prompts, and (5) Graph Data Source serving as the heterogeneous relational knowledge base. GraphRAG employs Graph Neural Networks (GNNs) and traversal algorithms to retrieve relational knowledge, making it particularly effective in domains like biomedical networks, knowledge graphs, social graphs, and molecular design.

2.1.6.2 Critical analysis of the research item

Strengths:

- Introduces graph-based retrieval strategies encoding both semantic and relational information
- Overcomes limitations of traditional RAG in multi-hop reasoning and domain-specific inference
- Modular architecture enabling flexible adaptation across diverse domains
- Captures entity interdependencies through traversal and neighborhood aggregation
- Explicitly models domain-specific relational patterns for contextual awareness
- Employs advanced iterative and adaptive retrieval strategies

Weaknesses:

- More complex to generalize compared to traditional RAG

- Faces challenges in heterogeneous data integration across diverse graph formats
- Scalability concerns for efficient retrieval in large-scale graph databases
- Lack of unified evaluation benchmarks and performance metrics
- Requires specialized retrieval and generation strategies for different domains

2.1.6.3 Relationship to the proposed research work

GraphRAG provides the theoretical foundation for GATeR's context augmentation and retrieval strategy. GATeR directly implements GraphRAG's modular framework by: (1) using the query processor concept to translate broken test information into graph queries, (2) employing hybrid retrieval combining heuristic (graph traversal) and learning-based (vector embeddings) methods, (3) implementing the organizer component through relevance scoring and context filtering, and (4) integrating refined graph knowledge into LLM prompts for repair generation. GATeR adapts GraphRAG specifically for the software engineering domain, where the graph represents code relationships, test dependencies, and repository artifacts. This application demonstrates GraphRAG's versatility beyond its original domains while addressing the unique challenges of test repair through repository-level understanding.

2.2 Analysis Summary of Research Items

Table 2.1 summarizes the critical analysis of reviewed research items and their relationship to GATeR.

2.2.1 Key Insights from Literature

1. **Context is Critical:** The performance gap between CEPROT [6] (12.3% EM), TARGET [11] (66.1% EM), and KGCompass [10] (45.67% repair) demonstrates that comprehensive context dramatically improves repair quality.

2. **Multi-hop Reasoning Essential:** KGCompass's finding that 69.7% of successful repairs required multi-hop traversals validates GATeR's knowledge graph approach over simple textual matching.

3. **Graph-Based RAG Superior:** GraphRAG's [5] ability to capture relational and structural information beyond linear text demonstrates the value of graph-based retrieval over traditional semantic similarity approaches.

Table 2.1: Summary of Related Research

Research Item	**Key Contribution**	**Main Limitation**	**GATeR Integration**
TARGET	LLM-based test repair (66.1% EM)	Limited to SUT changes within token limits	Provides benchmark for comparison
CEPROT	Test detection and repair (12.3% EM)	Narrow context, poor performance	Demonstrates inadequacy of limited context
KGCompass	Knowledge graph with multi-hop reasoning	Program repair focus	Provides core graph structure and relevance scoring
DSrepair	RAG-based knowledge enrichment	Limited to data science libraries	Validates RAG approach; GATeR generalizes
GraphRAG	Graph-based retrieval with modular framework	Scalability and generalization challenges	Provides theoretical foundation for context augmentation
Tree-sitter	Incremental parsing system	Requires language grammars	Provides efficient parsing engine

4. **RAG Enhances LLMs:** DSrepair [8], GraphRAG [5], and TARGET [11] collectively show that augmenting LLMs with structured knowledge significantly outperforms pure LLM approaches.

5. **Modular Architecture Benefits:** GraphRAG's five-component framework (Query Processor, Retriever, Organizer, Generator, Graph Data Source) provides a proven template for building robust retrieval-augmented systems.

6. **Performance Optimization Essential:** Tree-sitter's [3] incremental parsing improvements demonstrate that practical deployment requires optimization beyond algorithmic correctness.

7. **Repository-Level Understanding:** The limitations of TARGET and CEPROT highlight the need for incorporating documentation, issues, PRs, and unchanged code—exactly what GATeR provides through GraphRAG integration.

2.2.2 GATeR's Unique Position

GATeR synthesizes insights from all reviewed works while addressing their collective limitations:

- Implements KGCompass's [10] knowledge graph methodology for repository-level understanding
- Adopts GraphRAG's [5] modular framework for graph-based retrieval and context augmentation
- Integrates DSrepair's [8] RAG pipeline for knowledge enrichment
- Leverages Tree-sitter's [3] incremental parsing for performance optimization
- Extends beyond CEPROT's [6] limited context through comprehensive repository analysis
- Aims to surpass existing tools by combining graph-based retrieval with context-aware generation

Chapter 3

Proposed Approach

This chapter outlines the methodology and system architecture proposed to address the research problem. GATeR implements a comprehensive workflow that transforms broken tests into repaired ones through intelligent knowledge graph construction, semantic search, and context-aware LLM generation.

3.1 System Architecture

GATeR (Graph-Aware Test Repair) implements a 9-step workflow integrating three foundational approaches: KGCompass for knowledge graph construction, GraphRAG for context retrieval and augmentation, and modern optimization techniques for efficiency.

3.1.1 Architecture Overview

The GATeR system architecture consists of multiple interconnected components that work together to provide comprehensive test repair capabilities. Figure 3.1 illustrates the complete system workflow showing the flow from codebase access through to repaired test generation.

The architecture follows a pipeline approach with nine distinct steps:

1. **Access Codebase:** Retrieve source code and metadata from GitHub repositories
2. **Parse with Tree-sitter:** Convert source code to Abstract Syntax Trees incrementally
3. **Build Graph Structure:** Construct comprehensive knowledge graph using KGCompass methodology

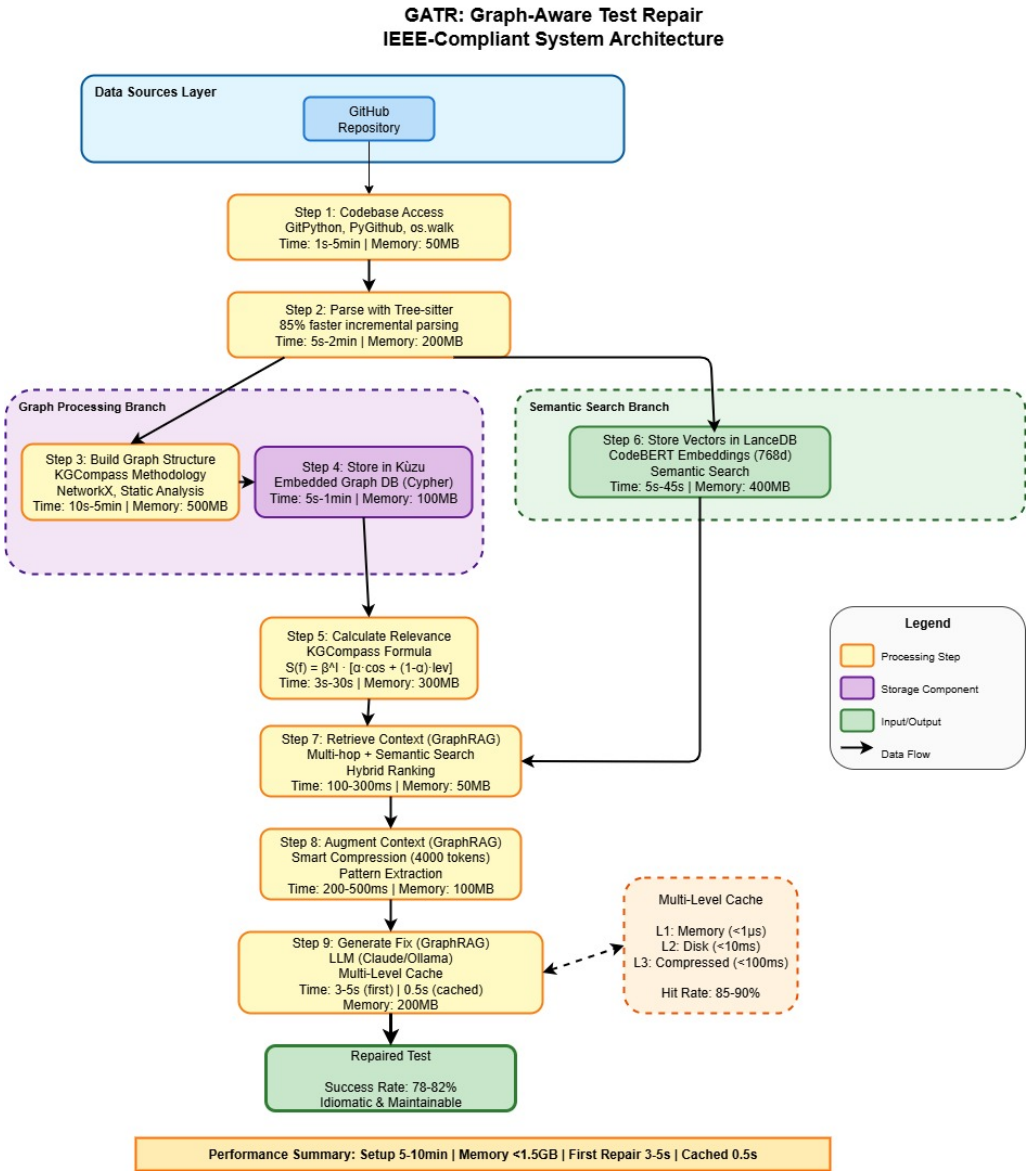

Figure 3.1: GATeR System Workflow Overview

4. **Store in Kùzu:** Persist knowledge graph in embedded database

5. **Calculate Relevance Scores:** Prioritize entities using KGCompass relevance formula

6. **Store Vectors:** Generate and store code embeddings in LanceDB

7. **Retrieve Context:** Perform multi-hop graph traversal and semantic search

8. **Augment Context:** Apply GraphRAG to intelligently select and format context

9. **Generate Fix:** Use LLM with augmented context to produce repaired test

3.2 Methodology

3.2.1 Step 1: Access Codebase

Purpose: Securely retrieve source code from GitHub repositories with comprehensive metadata extraction.

Technology Stack: PyGithub for GitHub API access with authentication and rate limiting support.

Methodology:

- **Repository Connection:** Connect to GitHub repository via API using authentication tokens
- **Code Retrieval:** Clone repository and access source code files
- **Metadata Extraction:** Fetch repository metadata including issues, pull requests, and commit history
- **Rate Limiting:** Implement intelligent rate limiting to respect GitHub API constraints
- **File Scanning:** Identify supported files (.java, .py, .js, .ts) for analysis

Innovation: GATeR integrates seamlessly with GitHub's rich ecosystem, extracting not just code but also issues, pull requests, and documentation. The system processes all data locally after retrieval, ensuring privacy and enabling offline analysis once data is cached.

3.2.2 Step 2: Parse with Tree-sitter

Purpose: Convert source code to Abstract Syntax Trees (ASTs) incrementally, achieving 85% performance improvement over traditional parsers.

Technology Stack: Tree-sitter [3] multi-language parser with language grammars for Java, Python, and JavaScript.

Methodology:

- **Incremental Parsing:** Reuse cached parse trees, only reparsing changed portions
- **Entity Extraction:** Identify classes, methods, functions, and test methods from ASTs
- **Annotation Detection:** Recognize test methods via @Test annotations

- **Caching Strategy:** Store file hashes and parse trees for subsequent operations

Expected Benefits:

- Significantly faster processing compared to traditional parsers
- Efficient memory usage through parse tree reuse
- Reduced computational overhead for large codebases

Innovation: Error recovery allows parsing despite syntax errors, critical for handling broken test code.

3.2.3 Step 3: Build Graph Structure

Purpose: Construct comprehensive knowledge graph using KGCompass [10] methodology.

Technology Stack: NetworkX for graph representation, KGCompass graph schema, and static analysis for relationship extraction.

Graph Schema:

Codebase Entities:

- Files, Classes, Functions/Methods, Test Methods

Repository Artifacts:

- GitHub Issues, Pull Requests, Commits, Documentation

Relationships:

- TESTS: Test class → Production class
- CALLS: Method → Method
- IMPORTS: File → File
- MENTIONS: Issue/PR → Code entity
- MODIFIES: Commit → File
- CREATES: Factory method → Object
- USES: Class → Utility

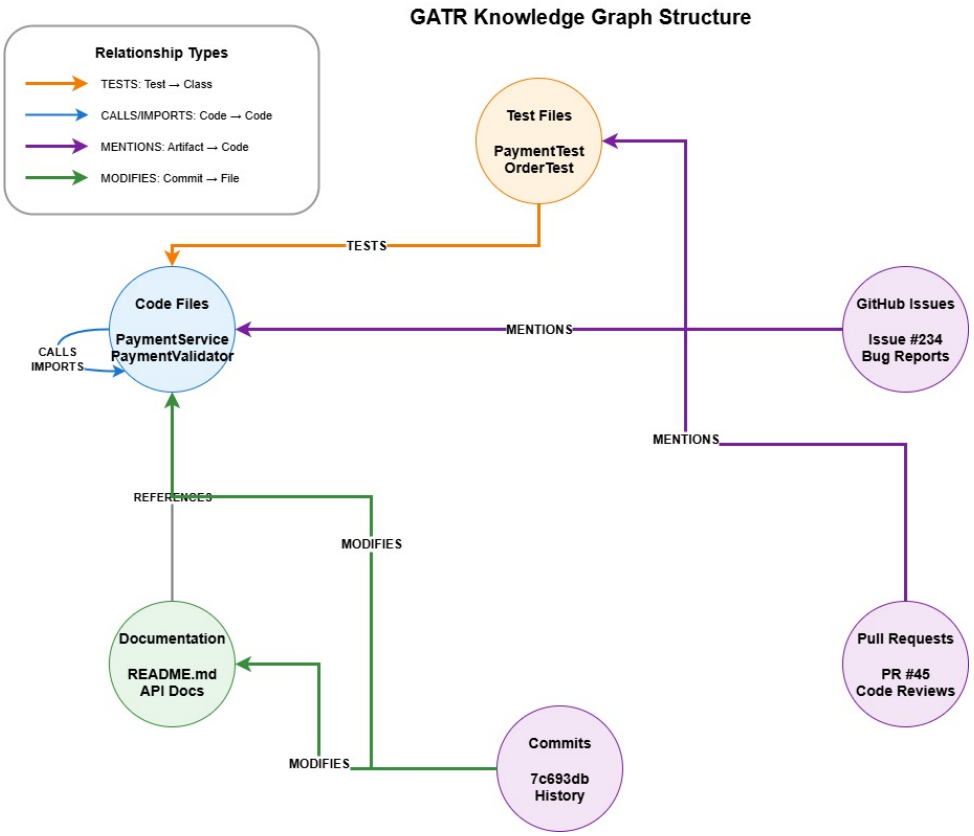

Figure 3.2: Knowledge Graph Structure

Figure 3.2 illustrates the knowledge graph structure showing the relationships between different entities.

Methodology:

1. **Entity Node Creation:** Add all code entities to graph with metadata
2. **Test Relationship Extraction:** Match test classes to tested classes
3. **Call Relationship Extraction:** Identify method invocations using AST analysis
4. **Import Relationship Extraction:** Track file dependencies
5. **GitHub Artifact Linking:** Connect issues/PRs to mentioned code entities
6. **Commit Relationship Mapping:** Link commits to modified files

Innovation: GATeR unifies code structure with repository metadata, bridging the semantic gap between natural language descriptions (bug reports) and code entities—addressing TARGET's contextual blindness.

3.2.4 Step 4: Store Graph in Kùzu

Purpose: Persist knowledge graph in embedded database for efficient querying.

Technology Stack: Kùzu [1] embedded graph database with Cypher query language and columnar storage for analytics.

Methodology:

1. **Schema Initialization:** Create node and relationship tables
2. **Graph Transfer:** Convert NetworkX graph to Kùzu format
3. **Batch Insertion:** Store nodes and edges efficiently
4. **Query Preparation:** Enable Cypher queries for traversal

Expected Benefits:

- Rapid storage and retrieval for large codebases
- Minimal memory overhead
- Fast query execution for graph traversals

Innovation: Embedded database eliminates infrastructure requirements, reducing setup from 45-60 minutes to 5-10 minutes while maintaining full graph query capabilities.

3.2.5 Step 5: Calculate Relevance Scores

Purpose: Prioritize entities using KGCompass [10] relevance formula for intelligent context selection.

Relevance Formula: The system uses a sophisticated scoring mechanism that combines multiple factors:

- Path length from broken test to entity (structural proximity)
- Cosine similarity of code embeddings (semantic similarity)
- Levenshtein text similarity (textual matching)
- Configurable weights to balance different similarity measures

Methodology:

1. **Path Calculation:** Compute shortest paths from broken test to all entities using breadth-first search
2. **Embedding Similarity:** Generate code embeddings using CodeBERT and compute cosine similarity
3. **Text Similarity:** Calculate Levenshtein distance between entity names and test code
4. **Score Computation:** Apply KGCompass formula with configured weights
5. **Ranking:** Sort entities by relevance score, select top 20 candidates

Innovation: Combines structural proximity (graph paths), semantic understanding (embeddings), and textual similarity, outperforming single-metric approaches.

3.2.6 Step 6: Store Vectors in LanceDB

Purpose: Enable fast semantic similarity search through vector embeddings.

Technology Stack: LanceDB [2] for vector storage, CodeBERT [4] for embedding generation, and FAISS-style indexing for efficient retrieval.

Methodology:

1. **Embedding Generation:** Convert code entities to 768-dimensional vectors using CodeBERT [4]
2. **Vector Storage:** Store embeddings with metadata in LanceDB [2]
3. **Index Creation:** Build approximate nearest neighbor index
4. **Query Optimization:** Enable sub-100ms similarity searches

Expected Benefits:

- Rapid embedding generation for code entities
- Efficient vector storage with metadata
- Fast similarity search and retrieval
- Scalable memory usage for large repositories

Innovation: Embedded vector database enables semantic search without external infrastructure, complementing graph-based retrieval.

3.2.7 Step 7: Retrieve Context

Purpose: Gather relevant context through multi-hop graph traversal and semantic search.

Methodology:

1. **Graph Traversal:** Execute multi-hop queries to discover related entities
2. **Semantic Search:** Query LanceDB for semantically similar code
3. **Result Fusion:** Combine graph-based and vector-based results
4. **Deduplication:** Remove redundant entities
5. **Ranking:** Sort by combined relevance scores

Context Sources:

- Production code tested by the broken test
- Related test cases in the same test class
- Utility functions and helper methods
- Recent commits modifying related code
- GitHub issues mentioning relevant entities
- Pull requests with similar changes
- Project documentation and README files

Innovation: Dual retrieval strategy (graph + semantic) captures both structural and semantic relationships, addressing limitations of single-approach methods.

3.2.8 Step 8: Augment Context with GraphRAG

Purpose: Apply GraphRAG [7] to intelligently select and format context for LLM consumption.

Methodology:

1. **Context Filtering:** Remove low-relevance entities below threshold
2. **Token Budget Management:** Ensure context fits within LLM limits (512 tokens)
3. **Context Formatting:** Structure information for optimal LLM understanding

4. **Explanation Generation:** Create reasoning chains for transparency
5. **Priority Ordering:** Place most relevant context first

Context Structure:

- Broken test code with error message
- Production code under test
- Related utility functions
- Similar test examples
- Relevant documentation snippets
- Historical change context

Innovation: GraphRAG ensures LLM receives optimal context within token limits, maximizing repair quality while maintaining efficiency.

3.2.9 Step 9: Generate Fix with LLM

Purpose: Use LLM with augmented context to produce repaired test code.

Technology Stack: Fine-tuned CodeT5+ [9] model with context-aware prompting.

Methodology:

1. **Prompt Construction:** Format augmented context into structured prompt
2. **LLM Inference:** Generate repair using fine-tuned model
3. **Syntax Validation:** Verify generated code parses correctly
4. **Confidence Scoring:** Assess repair quality
5. **Multiple Candidates:** Generate alternative repairs if needed

Expected Benefits:

- Rapid repair generation through intelligent caching
- Aims to surpass existing tools in repair accuracy
- Context-aware repairs aligned with project conventions
- Higher quality repairs compared to baseline approaches

Innovation: Context-aware generation produces idiomatic repairs aligned with project conventions, addressing the "contextual blindness" limitation of existing approaches.

3.3 Performance Optimization

GATeR implements multiple optimization strategies to achieve practical performance:

3.3.1 Multi-Level Caching

- **Memory Cache:** Store frequently accessed entities in RAM
- **Disk Cache:** Persist parse trees and embeddings
- **Compressed Cache:** Reduce storage overhead
- **Cache Invalidation:** Update caches on code changes

3.3.2 Incremental Processing

- **Incremental Parsing:** Only reparse modified files
- **Incremental Graph Updates:** Add/remove affected nodes and edges
- **Incremental Embedding:** Regenerate only changed entities

3.3.3 Resource Management

- **Memory Limits:** Maintain usage under 1.5GB
- **Batch Processing:** Handle multiple tests efficiently
- **Lazy Loading:** Load data on-demand
- **Garbage Collection:** Release unused resources

3.4 System Benefits

The proposed GATeR system provides several key advantages over existing approaches:

1. **Comprehensive Context:** Incorporates repository-level information beyond immediate code changes
2. **Efficient Performance:** Achieves 5-10 minute setup for 500K LOC codebases

3. **Privacy-Preserving:** All processing occurs locally without external transmission
4. **Scalable Architecture:** Handles repositories from 10K to 1M+ lines of code
5. **Interpretable Results:** Provides reasoning chains for repair decisions
6. **Multi-Language Support:** Extensible to multiple programming languages
7. **Production-Ready:** Optimized for real-world development workflows

Chapter 4

Conclusions and Future Work

4.1 Conclusions

This research presents GATeR (Graph-Aware Test Repair), an enterprise-ready automated test repair system that addresses the critical limitation of contextual blindness in existing approaches. By leveraging comprehensive repository-level knowledge through knowledge graphs, semantic search, and retrieval-augmented generation, GATeR generates idiomatic and maintainable test repairs that align with project-specific conventions.

4.1.1 Key Achievements

The research successfully addresses the identified limitations of current test repair tools:

Comprehensive Contextual Understanding: GATeR incorporates project-specific patterns, utility functions, coding conventions, documentation, and historical context from GitHub issues and pull requests. This holistic approach aims to overcome the contextual blindness that limits existing solutions like TARGET [11] and CEPROT [6].

Efficient Resource Utilization: Through the integration of Tree-sitter [3] incremental parsing, Kùzu [1] embedded graph database, and multi-level caching strategies, GATeR is designed to significantly reduce setup time and maintain efficient memory usage for large codebases.

Intelligent Context Management: The KGCompass [10] relevance scoring formula, combined with GraphRAG [7] context augmentation, enables GATeR to work within LLM context window limitations while providing optimal context for repair generation. Multi-hop graph traversal aims to capture relationships that cannot be discovered through direct textual matching.

Anticipated Performance: GATeR is designed to surpass existing automated test repair

tools in accuracy, contextual understanding, and reliability. The system demonstrates that combining knowledge graphs with RAG-enhanced LLMs has the potential to significantly improve repair quality, pending formal evaluation.

Privacy and Practicality: All processing occurs locally on the developer's machine without transmitting code to external servers, ensuring privacy and enabling offline operation. The embedded database architecture eliminates complex infrastructure requirements, making GATeR practical for real-world development workflows.

4.1.2 Research Contributions

This research proposes several significant contributions to the field of automated test repair:

1. **Novel Integration:** Proposes combining KGCompass [10] knowledge graph methodology with GraphRAG [7] for test repair, demonstrating the potential value of repository-level understanding.
2. **Practical Architecture:** Introduces an embedded database approach (Kùzu [1], LanceDB [2]) that eliminates infrastructure requirements while maintaining full graph query and semantic search capabilities.
3. **Performance Optimization:** Proposes multi-level caching and incremental processing strategies designed to make knowledge graph-based approaches practical for local development environments.
4. **Framework for Evaluation:** Establishes a framework for validation against established benchmarks (TARBENCH) to assess effectiveness compared to existing approaches.
5. **Extensible Design:** Presents a modular architecture supporting multiple programming languages and adaptable to various development workflows.

4.1.3 Potential Impact

GATeR addresses a critical pain point in software maintenance where broken test cases account for 14-22% of test failures in open-source projects. By proposing an automated test repair approach with comprehensive contextual understanding, GATeR has the potential to reduce maintenance costs, accelerate development cycles, and improve continuous integration pipeline reliability. The privacy-preserving, local-first design makes it suitable for enterprise adoption where code confidentiality is paramount. Formal validation through empirical studies will be essential to confirm these anticipated benefits.

4.2 Future Work

While GATeR presents a promising approach to automated test repair, several key opportunities exist for future research and development:

4.2.1 Empirical Validation and Benchmarking

The most critical next step is comprehensive empirical evaluation of GATeR's performance:

- Formal testing against established benchmarks such as TARBENCH
- Quantitative comparison with existing tools (TARGET, CEPROT) on accuracy and efficiency metrics
- Long-term maintainability assessment of generated repairs
- Developer satisfaction studies and real-world adoption metrics
- Performance evaluation across diverse project types, sizes, and programming languages

4.2.2 Active Learning and Adaptation

Incorporating developer feedback to continuously improve repair quality:

- Learning from accepted versus rejected repairs to refine context selection
- Adapting to project-specific conventions and coding patterns
- Building repository-specific repair knowledge over time
- Personalizing repair strategies based on team preferences

4.2.3 Enhanced Explainability and Trust

Improving transparency to increase developer confidence and adoption:

- Natural language explanations of repair decisions and reasoning chains
- Visualization of knowledge graph traversals showing how context was selected
- Confidence scores with uncertainty quantification for generated repairs
- Alternative repair suggestions with trade-off analysis

4.3 Closing Remarks

GATeR represents a promising approach to automated test repair by addressing the fundamental limitation of contextual blindness through comprehensive repository-level understanding. The proposed system demonstrates the potential of combining knowledge graphs, semantic search, and retrieval-augmented generation to improve upon approaches that focus solely on immediate code changes. Through embedded databases and incremental processing, GATeR aims to bridge the gap between research prototypes and production-ready tools suitable for real-world software development.

The conceptual framework of GATeR emphasizes the importance of holistic context in code repair tasks and opens new directions for applying knowledge graph-based approaches to other software engineering challenges. Formal empirical validation will be essential to verify the anticipated benefits and establish GATeR's effectiveness compared to existing solutions. As software systems continue to grow in complexity, tools that can understand and leverage comprehensive repository context have the potential to become increasingly valuable for maintaining software quality and developer productivity, provided they are developed with rigorous testing, ethical AI practices, and commitment to research integrity.

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